

Aobo Chen

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EDUCATION

Doctor of Philosophy in Computer Science

Expected Spring 2029

Iowa State University, Ames, IA

Advisor: Dr. Mengdi Huai

Master of Science in Computer Science

May 2025

Iowa State University, Ames, IA

Advisor: Dr. Mengdi Huai

Bachelor of Science in Computer Science

July 2023

Sichuan University, Chengdu, China

RESEARCH INTERESTS

My research focuses on data mining and machine learning. I am particularly interested in studying the security and privacy vulnerabilities of machine learning algorithms, with a focus on machine unlearning, uncertainty quantification, interpretability, and medical imaging applications.

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2024 - Present

Iowa State University, Ames, IA

- Conduct research on security and privacy vulnerabilities in machine learning systems, collaborating with supervisor, research scientists, and fellow graduate students on interdisciplinary projects.
- Study vulnerabilities of large reasoning models in machine unlearning by developing attack frameworks that induce incorrect final answers with convincing but misleading reasoning traces in both white-box and black-box settings.
- Investigate security and privacy risks in machine unlearning through a comprehensive study of attack surfaces, threat categories, countermeasures, and open challenges across the full unlearning lifecycle.
- Design efficient uncertainty quantification and explanation methods for medical image classification by developing approximate full conformal prediction and influential-sample analysis.

RESEARCH PUBLICATIONS

- **Aobo Chen***, Chenxu Zhao*, Chenglin Miao, and Mengdi Huai. "Towards Unveiling Vulnerabilities of Large Reasoning Models in the Context of the Right to Be Forgotten." The Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD 2026**), Hong Kong, China, June 2026.
- Chenxu Zhao, Wei Qian, **Aobo Chen**, and Mengdi Huai. "Membership Inference Attacks with False Discovery Rate Control." The 2025 IEEE/CVF International Conference on Computer Vision (**ICCV 2025**), Honolulu, Hawaii, USA, October 2025.
- **Aobo Chen***, Yangyi Li*, Chenxu Zhao, and Mengdi Huai. "A Survey of Security and Privacy Issues of Machine Unlearning." AI Magazine, 2025.

* indicates equal contribution

- **Aobo Chen**, Yangyi Li, Wei Qian, Kathryn Morse, Chenglin Miao, and Mengdi Huai. “Modeling and Understanding Uncertainty in Medical Image Classification.” The 27th International Conference on Medical Image Computing and Computer-Assisted Intervention (**MICCAI 2024**), Marrakesh, Morocco, October 2024.
- Yangyi Li*, **Aobo Chen***, Wei Qian, Chenxu Zhao, Divya Lidder, and Mengdi Huai. “Data Poisoning Attacks Against Conformal Prediction.” The 41st International Conference on Machine Learning (**ICML 2024**), Vienna, Austria, July 2024.
- Chenxu Zhao, Wei Qian, Yangyi Li, **Aobo Chen**, and Mengdi Huai. “Rethinking Adversarial Robustness in the Context of the Right to Be Forgotten.” The 41st International Conference on Machine Learning (**ICML 2024**), Vienna, Austria, July 2024.
- Wei Qian*, **Aobo Chen***, Chenxu Zhao, Yangyi Li, and Mengdi Huai. “Exploring Fairness in Educational Data Mining in the Context of the Right to be Forgotten.” ArXiv, May 2024.

Tutorials

- **Aobo Chen**, Wei Qian, Zheyuan Liu, Shagufta Mehnaz, Tianhao Wang, and Mengdi Huai. “Data Security and Privacy in Machine Unlearning: Recent Advances, Challenges, and Future Perspectives.” Tutorial at the 25th IEEE International Conference on Data Mining (**ICDM 2025**), Washington, DC, USA, November 2025.

Under Review

- **Aobo Chen**, Wei Qian, Chenglin Miao, and Mengdi Huai. “Towards Uncovering and Correcting Biases in Conformal Prediction.” The 19th European Conference on Computer Vision (**ECCV 2026**).
- **Aobo Chen**, Chenxu Zhao, Wei Qian, Zheming Cao, Tianhao Wang, and Mengdi Huai. “SUAD-Bench: Towards a Comprehensive Benchmark for Security Risks in Machine Unlearning under the Right to Be Forgotten.” The 19th European Conference on Computer Vision (**ECCV 2026**).
- Chenxu Zhao, Wei Qian, **Aobo Chen**, Jingquan Wang, Carl Yang, and Mengdi Huai. “LUSB: Formalizing and Benchmarking Unlearning Attacks and Defenses against Large Language Models.” The 43rd International Conference on Machine Learning (**ICML 2026**).

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug. 2023 - Dec. 2024

Iowa State University, Ames, IA

- Assisted in teaching COMS 230 Discrete Computational Structures and COMS 573 Machine Learning.
- Held office hours to assist students with course assignments and technical questions.

SELECTED MENTORSHIP

Graduate Research Mentor

May 2025 - Aug. 2025

Iowa State University, Ames, IA

- Mentored Zheming Cao, a graduate student in Statistics, on research exploring security risks in machine unlearning under the right to be forgotten.
- Provided guidance on experimental design, code implementation, and academic paper writing.

Undergraduate Research Mentor

Jan. 2024 - May 2024

Iowa State University, Ames, IA

- Mentored Kathryn Morse, an undergraduate student in Computer Science, on research in uncertainty quantification for medical image classification.
- Provided guidance on project development, literature review, and academic paper writing.

HONORS AND AWARDS

- Publication Award, Department of Computer Science, Iowa State University, 2025
- Publication Award, Department of Computer Science, Iowa State University, 2024
- Mathematical Contest in Modeling Honorable Mention, Sichuan University, 2021
- National Undergraduate Innovation and Entrepreneurship Training Program Project Completion Certificate, Sichuan University, 2021
- College Undergraduate Innovation and Entrepreneurship Training Program Project Completion Certificate, Sichuan University, 2021

PROFESSIONAL SERVICE

Reviewer and External Reviewer

- Reviewer for the Neural Information Processing Systems (NeurIPS), 2026
- Reviewer for the AAAI Conference on Artificial Intelligence (AAAI), 2026
- Reviewer for the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- Reviewer for the AAAI Conference on Artificial Intelligence (AAAI), 2025
- External reviewer for the Association for Computational Linguistics (ACL), 2026
- External reviewer for the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- External reviewer for the Network and Distributed System Security (NDSS) Symposium, 2026
- External reviewer for the International Conference on Learning Representations (ICLR), 2025

Other Service

- Teaching assistant, Summer Bootcamp on Deep Learning and Applications, 2024